

Value-of-Information Aware Active Task Assignment

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ABSTRACT

This paper discusses the problem of robust allocation of unmanned vehicles (UV) to targets with uncertainties. In particular, the team consists of heterogeneous vehicles with different exploration and exploitation abilities. A general framework is presented to model uncertainties in the planning problems, which goes beyond traditional Gaussian noise. Traditionally, exploration and exploitation are decoupled into two assignment problems are planned with un-correlated goals. The coupled planning method considered here assign exploration vehicles based on its potential influence of the exploitation. Furthermore, a fully decentralized algorithm, Consensus-Based Bundle Algorithm (CBBA), is used to implement the decoupled and coupled methods. CBBA can handle system dynamic constraints such as target distance, vehicle velocities, and has computation complexity polynomial to the number of vehicles and targets. The coupled method is shown to have improved planning performance in a simulated scenario with uncertainties about target classification.

1. INTRODUCTION

Future Unmanned Vehicle (UV) missions require high-level autonomous planning algorithms that are robust to environmental uncertainties and use information from communication with other UVs/sensors to improve performance. Example scenarios include target tracking/surveillance with environmental uncertainties. In many distributed planning scenarios, the uncertainty about the target is significant that the potential score that can be obtained by executing a task at the target is highly affected. In such situations, it is important to plan to utilize any available dedicated exploration vehicles to explore the target and improve confidence in target. One approach to tackle the resulting exploration-exploitation problem is to solve the exploration problem to reduce uncertainty first then assign vehicle to exploit. However, this approach does not prevent the exploration vehicles from exploring targets that have no mission value, and hence can lead to wasted resources. This paper presents an unified approach to plan heterogeneous UVs to explore targets as well as perform tasks at targets within a single universal planner that takes into account both the value of exploration and the reward obtained with task execution. We present a general framework for coupled exploration and planning in presence of categorical uncertainties in target type and score.

UV allocation problems is traditionally modeled as mathematical programming problems^{1,2}. In particular, robust optimization and stochastic programming techniques are used to deal with parametric uncertainty³⁻⁶. Authors have applied robust optimization techniques to UV planning problems^{2,7,8}. This framework assumes that the uncertainty about the environment and targets does not change during the mission. However, it is often the case that in a heterogeneous UV team, there are some UVs that are capable of exploring targets and reducing uncertainties. The problem of managing UVs/sensors to optimally reduce uncertainties in targets/environment is well-studied in sensor management literature⁹⁻¹³. However, in UV planning problems, the main goal is to maximize the mission scores in stead of purely reducing uncertainty. Therefore UV assignments only based on uncertainty reduction can potentially lead to resource waste on highly uncertain but low rewarding targets.

There is relatively limited work that considers both uncertainty and potential score at the same time while planning. Bertuccelli proposed a planning algorithm based on integer programming^{2,14} that establishes a coupling between the exploration and exploitation of targets by a team of heterogeneous UVs. However, that work is limited only to uncertainties that have a Gaussian distribution of the potential target score. While Gaussian

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distribution is a good model of uncertainty for measurement noise of continuous states such as positions, orientations and velocities, it cannot well represent discrete uncertainties such as the uncertainty in the category of the target. Figure 1 depicts a typical situation in which UVs need to be assigned to targets that fall into the category of tank, truck, or a car. The uncertainty in this case is the UVs prior guess on which target falls into what category before any measurements are taken. In this situation, Gaussian noise would not be the best descriptor of the uncertainty.

This paper presents an active planning architecture that can handle uncertainties in the categorical family. In particular, we seek an optimal solution to assign UVs in a heterogeneous team of Unmanned Tasking Vehicles (UTVs) and Unmanned Exploration Vehicles (UEVs) to targets in presence of uncertainty about the target type (and hence the score). We develop a framework that couples both exploitation action by UTVs and exploration action by UEVs such that the expected mission score is maximized. Our framework is designed specifically for discrete uncertainties, and hence goes beyond previous work that deals only with Gaussian uncertainties². Furthermore, we implement our approach with a Consensus-Based Bundle Algorithm (CBBA) to incorporate system dynamics such as agent velocities, and duration within which an action must be taken. Our framework is fully decentralized, and the computational complexity of obtaining a solution is polynomial in the number of targets and vehicles instead of exponential or combinatorial. Therefore it can scale well with number of vehicles and targets in distributed applications.

This paper is organized as follows: section 2 introduces planning under uncertainty; section 3 discusses how exploration abilities can help improve the performance; section 4 builds our approach in the decentralized CBBA framework; and simulation results are provided in section 5.

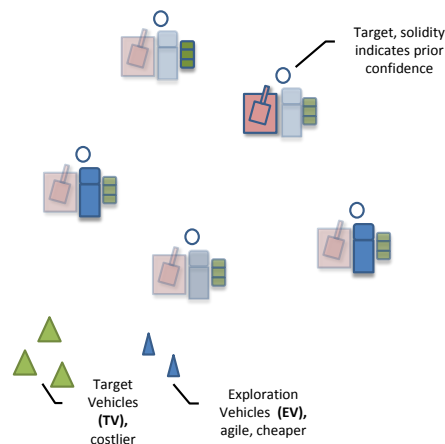


Figure 1. An example scenario of heterogeneous UV allocation

2. ROBUST PLANNING

In a typical UV target-association problem, the goal is to allocate a team of n unmanned vehicles (UVs) to N targets with the objective of maximizing the scores. The team accrues scores associated with the targets to which UVs are assigned to. If no targets are assigned, then it receives a score of 0. This problem becomes challenging when the scores are uncertain.

At time k , let vector $\mathbf{C}_k = [C_{1,k} \cdots C_{N,k}]^T$ denotes the score associated with N individual targets. Since the score is uncertain, $\mathbf{C}_k$ is a vector of random variables instead of deterministic values. Binary variable vector $\mathbf{x}_k = [x_{1,k} \cdots x_{N,k}]^T$ where $x_{i,k} \in \{0, 1\}$, denotes whether there is a UTV assigned to each task i at time k .

Further let $f(C) \in \mathcal{R}$ denote a functional mapping from the probability distribution of score C to a real number, then the planning problem can be recast as a mathematical programming problem:

$$\begin{aligned} \max \quad & \sum_{i=1}^N f(C_{i,k}) x_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \quad (1)$$

Score function $f(C)$ should capture two basic properties of C , the potential score and the uncertainty in it. Higher score and lower uncertainty should lead to larger $f(C)$. There are various ways of picking $f(C)$, and different $f(C)$ leads to different perspectives on potential score and uncertainty.

For example, the basic stochastic programming formulation uses expected score $f(C) = \mathbb{E}(C)$ ³:

$$\begin{aligned} \max \quad & \sum_{i=1}^N \mathbb{E}(C_{i,k}) x_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \quad (2)$$

This model only incorporates first moment information, which can potentially assign UTVs to targets with large uncertainty.

When C is bounded, $f(C)$ can account for uncertainty by taking the worst case value $f(C) = \min C$:¹⁵

$$\begin{aligned} \max_x \quad & \sum_{i=1}^N \left(\min_{C_{i,k}} C_{i,k} \right) x_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \quad (3)$$

This model can lead to very conservative plans because it is based on the worst case scenario.

Bertucci² proposed a planning formulation that can make a tradeoff between expected score and uncertainty. He assumes the distribution of score is Gaussian: $C \sim \mathcal{N}(\bar{c}, \sigma^2)$, and score function is $f(C) = \bar{c} - \mu\sigma$ in his framework. Compared to the expected value formulation (2) and the worst case formulation (3), this formulation strikes a balance between the expected score $\bar{c}$ and the uncertainty σ by picking different μ .

$$\begin{aligned} \max_x \quad & (\bar{c}_{i,k} - \mu\sigma_{i,k}) x_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \quad (4)$$

However, the assumption of Gaussian could be limiting in many cases. For example, when the uncertainty of the score comes from the classification of the target as shown in Figure 1, a Gaussian assumption fails to capture the uncertainty features.

The chance-constrained approach is another way to formulate robust planning problems^{16–19}. This approach guarantees the worst case score but allows a certain risk. As shown below, for each target i , the planner maximizes the worst case score $\hat{c}_{i,k}$, but allows a risk ϵ that the score is below $\hat{c}_{i,k}$.

$$\begin{aligned} \max_x \quad & \sum_{i=1}^N \hat{c}_{i,k} x_{i,k} \\ \text{s.t.} \quad & \mathbf{P}(C_{i,k} < \hat{c}_{i,k}) \leq \epsilon \\ & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \quad (5)$$

By picking different risk threshold ϵ , this algorithm can also strike a balance between the potential score and uncertainty in the target, and it does not constrain the form of uncertainties.

3. VALUE OF INFORMATION AWARE PLANNING

The formulations mentioned in Section 2 does not account for the case where vehicles are heterogeneous. When dedicated exploration vehicles exist, the team performance can be improved by assigning them to reduce the uncertainty in the scores. This section discusses how the presence of UEVs changes the optimal planning problem.

3.1 Value of Information Aware Exploration

UEVs are modeled by the procedure of taking measurements. Let $C_{i,k}$ denote the score distribution of target i at time k . If a UEV takes a measurement $z_{i,k}$ of this target, then the posterior score distribution, or the score distribution at time instant $k+1$ is the following by Bayes law:

$$\mathbf{P}(C_{i,k+1}) = \mathbf{P}(C_{i,k}|z_{i,k}) = \frac{\mathbf{P}(z_{i,k}|C_{i,k})\mathbf{P}(C_{i,k})}{\int \mathbf{P}(z_{i,k}|C_{i,k})d\mathbf{P}(C_{i,k})} \quad (6)$$

In general, $\mathbf{C}_{i,k+1}$ is a function of $z_{i,k}$, therefore $f(C_{i,k+1})$ is also a function of $z_{i,k}$. During the planning stage, $z_{i,k}$ is not available since UEV is not assigned to the target yet. A widely used technique in sensor management literature⁹⁻¹³ is to consider all possible outcomes of $z_{i,k}$ and compute expected $f(\cdot)$ on $z_{i,k}$, denoted as $\mathbb{E}f(C_{i,k+1})$:

$$\mathbb{E}f(C_{i,k+1}) = \mathbb{E}[f(C_{i,k}|z_{i,k})] = \int f(C_{i,k}|z_{i,k})d\mathbf{P}(z_{i,k}) \quad (7)$$

where $\mathbf{P}(z_{i,k}) = \int \mathbf{P}(z_{i,k}|C_{i,k})d\mathbf{P}(C_{i,k})$. Probability $\mathbf{P}(z_{i,k}|C_{i,k})$ represents the measurement model, it gives the likelihood of observing score $z_{i,k}$ when the true score is $C_{i,k}$. For example, if the measurement has Gaussian noise, $\mathbf{P}(z_{i,k}|C_{i,k}) \sim \mathcal{N}(C_{i,k}, \sigma^2)$.

Therefore, if a UEV is assigned to a target, the potential score for a UTV to do a task at this target will change from $f(C_{i,k})$ to $\mathbb{E}f(C_{i,k+1})$. The value of information obtained by assigning a UEV to this target is then the increment in the score function: $\mathbb{E}f(C_{i,k+1}) - f(C_{i,k})$. This notion of value of potential measurement actions can be used in the planning formulation to account for the ability of UEVs to take measurements.

3.2 Action-Exploration Decoupled Planning

Traditionally UTVs and UEVs are allocated separately with different score functions, then team assignments are got simply combining the UEV and UTV assignments.

Assume there are n UTVs and m UEVs. Binary variable vectors $\mathbf{x}_k = [x_{1,k} \cdots x_{N,k}]^T$, $\mathbf{y}_k = [y_{1,k} \cdots y_{N,k}]^T$ denote whether a UTV or a UEV is assigned to each of N targets at time k . Denote the possible measurements of each target at k as $\mathbf{z}_k = [z_{1,k} \cdots z_{N,k}]^T$.

The assignments of UTVs can be formed as the following robust planning problem (8):

$$\begin{aligned} \max \quad & \sum_{i=1}^N f(C_{i,k})x_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0,1\} \end{aligned} \quad (8)$$

The UEVs are allocated by solving another planning problem, with a different score function $g(C_{i,k})$ (9). Function $g(C_{i,k})$ should capture the uncertainty reduction of target i by exploring it.

$$\begin{aligned} \max \quad & \sum_{i=1}^N g(C_{i,k})y_{i,k} \\ \text{s.t.} \quad & \sum_{i=1}^N y_{i,k} \leq m \quad y_{i,k} \in \{0,1\} \end{aligned} \quad (9)$$

Equation (8) and Equation (9) together give a complete plan of assignments of UTVs and UEVs.

3.3 Action-Exploration Coupled Planning

In the decoupled algorithm, UTVs and UEVs are allocated separately. In particular, UTVs are assigned to targets with higher score functions $f(C_{i,k})$ while UEVs are assigned to targets with bigger uncertainty reduction $g(C_{i,k})$. It is often the case that these two targets sets do not have much overlap. If this is the case, UTVs do not benefit much from the exploration as UEVs are assigned to reduce uncertainty of other targets. Bertuccelli proposed a method to couple the assignments of UTVs and UEVs in a universal planner². However, his approach is only limited to Gaussian uncertainties. A similar approach is proposed here, which can be easily extended to categorical distributions:

$$\begin{aligned} \max \quad & \sum_{i=1}^N \mathbb{E}f(C_{i,k+1})x_{i,k}y_{i,k} + f(C_{i,k})x_{i,k}(1 - y_{i,k}) \\ \text{s.t.} \quad & \sum_{i=1}^N y_{i,k} \leq m, \quad y_{i,k} \in \{0, 1\} \\ & \sum_{i=1}^N x_{i,k} \leq n, \quad x_{i,k} \in \{0, 1\} \end{aligned} \tag{10}$$

In this coupled planning framework, the score function for a UTV to perform some task at target i is $f(C_{i,k})$ if no UEV is assigned to target i ($y_{i,k} = 0$), and is $\mathbb{E}f(C_{i,k+1})$ if a UEV is assigned to target i ($y_{i,k} = 1$). There is no separate score for UEVs. UEVs can add value only when the target will be assigned with a UTV. Therefore, this algorithm tends to assign UEVs to targets that will be assigned with UTVs at the same time. The cooperation can make better use of the team's heterogeneous abilities.

4. DISTRIBUTED PLANNING WITH SYSTEM DYNAMICS

The frameworks discussed in Section 3 are centralized and assume that the vehicle can get to the target immediately once being assigned to it. However, the system is often dynamic and there are physical constraints such as positions and task durations of targets, positions velocities of vehicles. These dynamics and constraints can effect the assignments results. For example, in order for the UTV to benefit from UEV's exploration at a target, the UEV should finish exploration before the corresponding UTV arrives at that target. Mathematical programming based approaches cannot deal with these constraints easily, because the increase in number of vehicles and tasks in dynamics systems leads to combinational increase in candidate solutions and becomes intractable quickly. It is also beneficial for the framework to be decentralized to get benefits as robustness and scalability. In this section, a polynomial decentralized algorithm, Coupled-Constraint Consensus-Based Bundle Algorithm (CCBBA) is introduced to decentralize the planning approach and handle system dynamics.

4.1 Coupled-Constraint Consensus-Based Bundle Algorithm (CCBBA)

Consensus-Based Bundle Algorithm (CBBA) is a distributed agreement protocol where agents bid on tasks to decide the assignments. It can well handle system dynamics, and provides provably good approximate solutions for multi-agent multi-task allocation problems over networks of agents^{20,21}.

CBBA consists of iterations between two phases: a bundle building phase where each vehicle greedily generates an ordered bundle of tasks, and a consensus phase where conflicting assignments are identified and resolved through local communication between neighboring agents. There are several core features of CBBA that can be exploited to develop an efficient planning mechanism for heterogeneous teams. First, CBBA is a decentralized decision architecture, which is a necessity for planning over large teams due to the increasing communication and computation overhead required for centralized planning. Second, the complexity of CBBA is polynomial in the number of tasks and vehicles. Third, various design objectives, agent models, and constraints can be incorporated by defining appropriate scoring functions.

Coupled-Constraint CBBA is built upon CBBA, which can further handle logical and temporal constraints between tasks²². CCBBA partition tasks into sub-groups, defined as activity, where logical and temporal constraints are defined upon.

For this paper there will be 3 types of constraints used by CCBBA for creating assignments. 1) mutex (two tasks cannot be assigned at the same time), 2) required dependencies (some tasks need to be assigned before other tasks are “released” and available for bidding), 3) temporal constraints (where timing constraints are required between tasks, in this work only the order with a slight buffer is required). Each of these can easily be encoded into the greedy bundle building phase for which tasks can be added next. See corresponding papers^{22,23} for details of how this is done.

4.2 Value of Information Aware Planning based on CCBBA

Each target i is defined by an activity $\mathcal{A}_i$. An activity consists of several tasks that represent different actions UVs can perform at the target. A task is defined by Table 1. In order to encourage vehicles to select shorter paths, the score function is exponentially discounted with time, with discount factor λ .

UVs are defined as in Table 2. A UV can pick some task into its bundle only when the UV’s type matches the task’s type. When a UV starts some task, it becomes occupied and the occupied time is specified as *availability*. UEVs are assumed more agile than UTVs, so the velocity of UEVs is higher than that of UTVs.

Table 1. Structure of Task

<i>ID</i>	uniquely identify task
<i>Type</i>	{Explore, Tasking}, match UVs
<i>Position</i>	position of target
<i>Duration</i>	duration of task at this target
<i>score</i>	score of task
λ	score decay rate of task

Table 2. Structure of UV

<i>ID</i>	uniquely identify UV
<i>Type</i>	{UEV, UTV}, match tasks
<i>Position</i>	current position of UV
<i>Velocity</i>	velocity of UV
<i>Availability</i>	Occupied time of UV

As shown in Table 3, in the decoupled planning approach, each activity has 2 tasks, *explore* and *tasking*. Notation t_e and t_t represent the time for exploring a target and tasking a target. Tasking time t_t is later than exploration time t_e . The score for exploring the target is uncertainty reduction function $g(C_i)$ while the score for tasking at a target is the prior score function $f(C_i)$.

In the coupled planning approach, each activity has 3 tasks: exploring a target: *explore*, tasking before exploring the target: *tasking_before*, and tasking after exploring: *tasking_after*. The score for *tasking_before* is the prior score $f(C_i)$, the score for *tasking_after* is the expected posterior score $\mathbb{E}f(C_{i,k+1})$. The score for *explore* is the increment in score $\mu(\mathbb{E}f(C_{i,k+1}) - f(C_{i,k}))$, multiplier μ controls the relative importance of exploration to the overall team score. The logical constraints include *tasking_before* and *tasking_after* are mutually exclusive; *tasking_after* depends on *explore*. The temporal constraint is that *explore* must lead *tasking_after* by at least t_e . The logical and temporal constraints are defined within activities as shown in Table 5

Table 3. Activity $\mathcal{A}_i$ in Decoupled Approach

explore	<i>Type</i> Explore <i>Duration</i> t_e <i>score</i> $g(C_i) - f(C_i)$
tasking	<i>Type</i> Tasking <i>Duration</i> t_t <i>score</i> $f(C_i)$

Table 4. Activity $\mathcal{A}_i$ in Coupled Approach

explore	<i>Type</i> Explore <i>Duration</i> t_e <i>score</i> $\mu(\mathbb{E}f(C_{i,k+1}) - f(C_i))$
tasking_before	<i>Type</i> Tasking <i>Duration</i> t_t <i>score</i> $\mathbb{E}f(C_{i,k+1})$
tasking_after	<i>Type</i> Tasking <i>Duration</i> t_t <i>score</i> $f(C_i)$

5. SIMULATION

In this section, we show the performance by implement the framework in Section 4 to an assignment problem with categorical score uncertainties.

Table 5. Constraints of Tasks in an Activity

temporal constraint	explore before tasking
logic constraint	tasking_before, tasking_after are mutual exclusive

5.1 Simulation Setup

The score uncertainty of a target is captured by a categorical distribution: score C of a target takes value in a finite set associated with K categories of targets: $[c^{(1)}, c^{(2)}, \dots, c^{(K)}]$. For example, when the uncertainty comes from classification of a target, the score can be modeled by a categorical distribution.

At time k , the score function $f(C_{i,k})$ is designed in such a way that it guarantees the worst case score $\hat{c}_{i,k}$ but allows a failure chance ϵ :¹⁹

$$f(C_{i,k}) = \hat{c}_{i,k} \quad \text{s.t.} \quad \mathbf{P}(C_{i,k} \leq \hat{c}_{i,k}) \leq \epsilon \quad (11)$$

UEVs have false detections with probability $1 - \gamma$: if the true score of the target is $c_{i,k}$, the UEV can get the right classification and the correct value $z_{i,k} = c_{i,k}$ with probability γ ; but it can also get the incorrect values $z_{i,k} \neq c_{i,k}$, with equal probability $\frac{1-\gamma}{K-1}$.

$$\mathbf{P}(z_{i,k}|c_{i,k}) = \begin{cases} \gamma & \text{if } z_{i,k} = c_{i,k} \\ \frac{1-\gamma}{K-1} & \text{otherwise} \end{cases} \quad (12)$$

A conjugate prior is used to simplify the computation of posterior score distributions. The conjugate prior for a categorical distribution is Dirichlet:

$$C_{i,k} \sim \text{Dir}(\alpha_{i,k}), \quad \alpha_k = [\alpha_{i,k}^{(1)}, \alpha_{i,k}^{(2)}, \dots, \alpha_{i,k}^{(K)}]^T \quad (13)$$

At time k , where N measurements $\mathbf{z}_{i,k}$ are observed and occurrence of each category is $N^{(1)}, N^{(2)}, \dots, N^{(K)}$, the posterior will be:

$$C_{i,k+1} \sim \text{Dir}(\alpha_{i,k+1}), \quad \alpha_{i,k+1} = [\alpha_{i,k}^{(1)} + N^{(1)}, \alpha_{i,k}^{(2)} + N^{(2)}, \alpha_{i,k}^{(K)} + N^{(K)}] \quad (14)$$

Therefore the expected posterior score function is

$$\begin{aligned} \mathbb{E}f(C_{i,k+1}) &= \mathbb{E}_{\mathbf{z}_{i,k}}(\hat{c}_{i,k+1}) = \sum_{i=1}^K \mathbf{P}(\mathbf{z}_{i,k}) \hat{c}_{i,k+1} \\ \mathbf{P}(\mathbf{z}_{i,k}) &= \sum_{j=1}^K \mathbf{P}(\mathbf{z}_{i,k}|c^{(j)}) \mathbf{P}(c^{(j)}) \end{aligned} \quad (15)$$

where $\mathbf{P}(c^{(j)})$ is given by Dirichlet prior (13), and $\mathbf{z}_{i,k}$ is given by the measurement model (12).

5.2 Planning Results

Figure 2 and Figure 3 show an example of the decoupled and coupled planning results. There are 10 targets in this map, each can be one of 3 classifications with score 10, 30 or 100. The histogram besides each circle shows the prior belief of classification of each target. Green triangles represent UTVs and blue triangles represent UEVs, and the speed of UEVs is twice as much as that of UTVs. Green and blue arrows represent their trajectories respectively. It can be seen that in the decoupled approach, UTVs are assigned to target based on highest score and UEVs are assigned to targets based on score increment. It is probable that they will end up heading to different sets of targets; if this happens, then the exploration by UEVs does not help reduce the uncertainties in targets assigned to UTVs. On the other hand, in the coupled case, all the UEVs are paired up with the UTVs, so UEVs can always help to reduce the uncertainty of tasks assigned to UTVs.

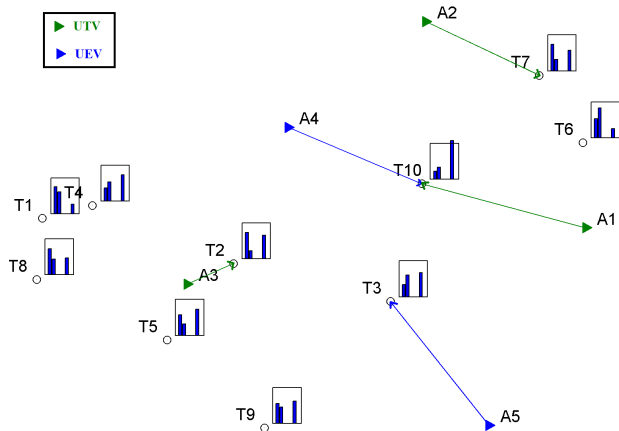


Figure 2. Decoupled allocation result. UTVs and UEVs can possibly go to different targets, in which case UEVs do not fully help UTVs to reduce uncertainty

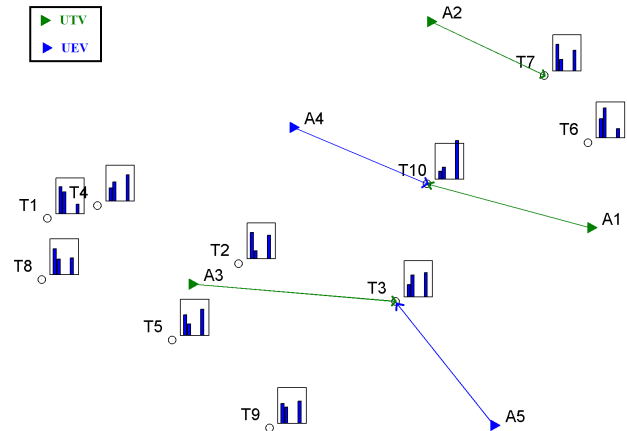


Figure 3. Coupled allocation result. UEVs group with UTVs, therefore UEVs can always help UTVs reduce uncertainty

Figure 4 and 5 shows statistics of 100 Monte Carlo simulations of different planning approaches. In each simulation, 30 tasks are randomly generated. The team has 3 UEVs and 5 UTVs, each takes 3 takes in their bundles and starts at random locations. UEV's speed is 2 times as that of UTV. When a UEV arrives at a task, it will spend 0.1 second to take 10 observations of the target, while UTVs need 1 second to finish the task at the target. Three different approaches are run to generate plans: in the first approach, only UTVs are assigned to tasks; the second is the decoupled approach which assigns UTVs and UEVs independently, and the last is the coupled approach. The score obtained by all UTVs and the expected running time to finish all the selected tasks are compared in figure 4 and 5

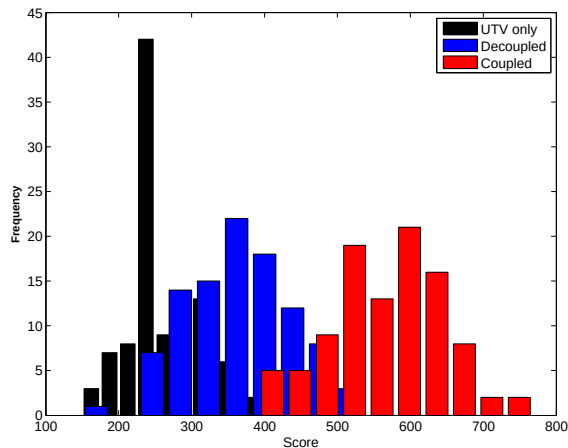


Figure 4. Histogram of score. Decoupled approach can overall get higher scores than UTV only case. Coupled approach gets even higher scores than decoupled approach.

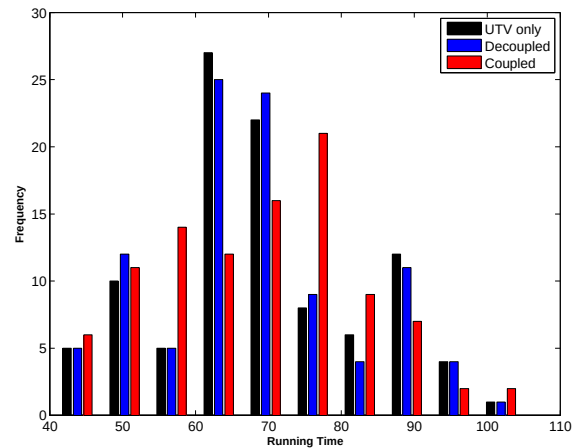


Figure 5. Histogram of expected running time. The plan generated by decoupled approach has similar running time as that generated by UTV only approach. But plan generated by coupled approach takes longer on average than that generated by decoupled approach.

In case only UTVs are assigned, the team does not utilize exploration ability to reduce uncertainty, so the score obtained by UTVs are among the lowest. In decouple approach, the plan of UEVs and UTVs are generated separately, the UTV can get higher score when a UEV happens to be assigned to explore that target before UTV gets there. The decoupled approach is always better than the UTV only case in terms of the scores obtained. In the coupled approach, UTVs and UEVs cooperate to explore as well as do tasks at targets, hence UEVs can always help UTVs to get higher score. Therefore, the coupled approach outperforms the decoupled approach in

terms of mission scores.

It is noticed that in the coupled approach, UTVs rely on the targets being explored by UEVs first so that it can decide whether or not to execute the task to obtain the score. Therefore, it is possible that some UTVs will wait for a UEV to explore a target before doing the task at that target, so the average time of coupled approach is observed to be longer than decoupled case.

5.3 Execution Result

In the previous section we presented the planning result before the mission was executed. In this section, we present the results after the mission is executed using the generated plans by simulating measurements of targets for UEVs. When a UEV explores a target, we assume that the likelihood of misclassification be 0.1, and it gets other wrong classifications with equal probability. Scenarios with different number of targets are considered. For each scenario, 100 Monte Carlo simulations are run. The performance of the three approaches used is compared in Figure 6

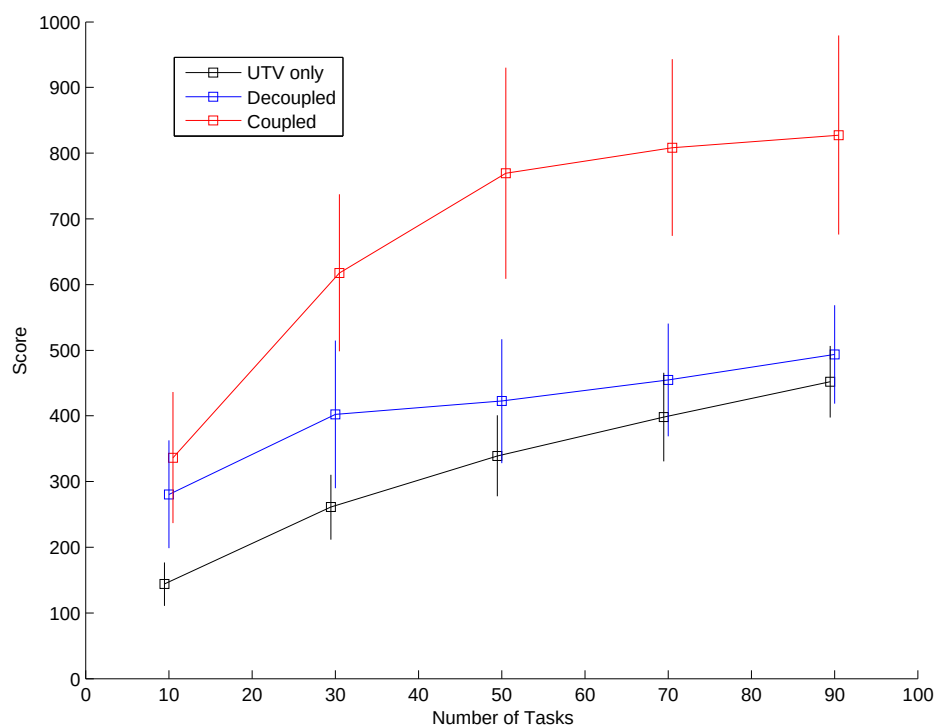


Figure 6. Statistics of 3 approaches with different number of targets. Coupled approach gets the highest score. Performance of decoupled approach is in between coupled and UTV only approaches. When there are less targets, performance of decoupled approach is closer to coupled approach. When there are more targets, performance of decoupled approach is closer to UTV only case.

The squares show the average score obtained by different approaches with different number of targets. The vertical lines associated with each square represents the standard deviation of each target amount. It can be seen that the coupled approach is able to get higher scores than others, as it utilizes the heterogeneous abilities of the team and encourages cooperation between them. When there are less targets, there is a bigger chance UEVs happen to pair with UTVs, therefore decoupled is closer to coupled. When there are more targets, it is harder for UEVs to pick the same targets as UTVs in decoupled approach, performance of decoupled approach is closer to UTV only case.

6. CONCLUSION

This paper discussed how to allocate heterogeneous Unmanned Vehicles (UVs) to targets with uncertainties. More specifically, we presented a general framework to couple the assignments of Unmanned Target Vehicles (UTVs) and Unmanned Exploration Vehicles (UEVs) to obtain better team performance. This approach can deal with various uncertainty forms, and thus goes beyond traditional Gaussian scenarios considered in previous work. Furthermore, we integrated this approach with the decentralized decision making Consensus-Based Bundle Algorithm (CBBA). This algorithm can handle dynamical constraints on the system, such as target locations, vehicle velocities, task durations etc.. CBBA also enabled us to fully decentralize the planning procedure, and thus, our approach scales well with number of vehicles, targets and target classifications. Stochastic simulation results showed that the coupled exploration-exploitation planning approach presented in this paper obtains better planning performance by encouraging cooperation between heterogeneous vehicles. Hence, it is possible to apply this approach to autonomous planning problems that are concerned with agents with heterogeneous capabilities.

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